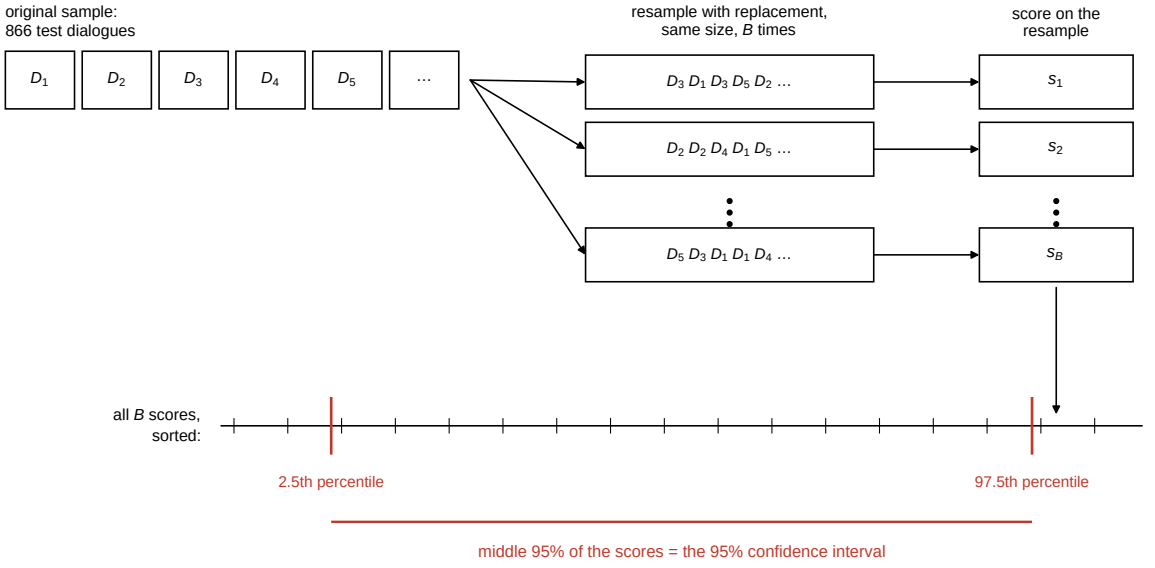
(a) Bootstrap: a confidence interval without formulas



Whole dialogues are resampled, not single markers, because the markers of one dialogue are not independent.

(b) McNemar's test: compare two systems on the same items

	B right	B wrong
A right	both right (ignored)	*b*: only A right
A wrong	*c*: only B right	both wrong (ignored)

If A and B are equally good,
b and *c* should be about equal.

$$\chi^2 = \frac{(b-c)^2}{b+c}$$

A large χ^2 gives a small *p*-value:
one system is really better on these items.